

Samuel Yuan

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EDUCATION

Carnegie Mellon University
B.S in Computer Science

Pittsburgh, PA
May 2028

- **Activities:** Tartan Scholars, Men's Club Ultimate Frisbee (Treasurer)
- **Relevant Coursework:** 15-418 Parallel Computer Architecture and Programming, 07-280 AI and ML, 18-213 Introduction to Computer Systems, 15-210 Parallel and Sequential DSA, 15-251 Theoretical Computer Science

EXPERIENCE

SailLoft
Software Development Intern

Chapel Hill, NC
Jun 2025– Aug 2025

- Designed and implemented Bazel-built plugins and extensions for Gerrit and GitHub in JavaScript to integrate developer workflow and testing suite into frontend UI, leading continuous integration tryjob management to make software testing more transparent and reducing turnaround time for developers
- Engineered webhook infrastructure with RESTful GET and POST APIs, enabling automated test triggers, monitoring, and real-time feedback in review interfaces for performance and stability assessment of enterprise codebase

PROJECTS

Parallel VLSI Wire Router | C++, OpenMP

March 2026

- Built a parallel VLSI routing system that computes wire paths across a chip grid while minimizing congestion costs
- Parallelized routing computations using OpenMP, distributing wire routing tasks across CPU cores
- Implemented cost-based grid routing and conflict resolution to improve routing efficiency
- Analyzed parallel speedup and scalability across varying input sizes and thread counts

CUDA Parallel Circle Renderer | C++, CUDA

Feb 2026

- Built a GPU-based circle renderer using CUDA that parallelizes pixel shading across thousands of threads
- Implemented CUDA kernels for circle rasterization and shading, computing pixel coverage and color blending
- Optimized for performance using coalesced memory accesses and shared memory, reducing global memory traffic
- Benchmarked against a CPU implementation and demonstrated substantial runtime improvement

Constructive Logic Theorem Prover | Standard ML, Prolog

Nov 2025

- Built a propositional logic theorem prover implementing proof search over the G4ip sequent calculus for SML and Prolog
- Implemented inference rules to determine derivability of logical propositions.
- Used continuation-passing style to implement efficient backtracking search in the proof engine.
- Applied invertible rule prioritization to reduce branching and improve proof search performance

HTTP Proxy & Web Caching | C

July 2025

- Developed a multi-threaded web proxy in C to handle HTTP/1.0 requests with concurrent client support
- Implemented LRU-based caching mechanism, reducing redundant network requests, improving response time
- Optimized throughput by managing concurrency with thread synchronization and efficient I/O buffering

Dynamic Memory Allocator | C

June 2025

- Constructed a malloc package in C with a segregated explicit free list to manage dynamic memory efficiently
- Implemented block splitting, coalescing, and best-fit search, balancing space utilization with speed
- Optimized memory usage with footerless allocated blocks and metadata bit-packing, achieving 74.4% utilization

Fantasy Runner Game | Python, pygame

Nov 2024

- Created a 2D auto-scrolling platformer game in Python, featuring randomized terrain and parallax background effects
- Implemented collision detection and sprite animation, enhancing gameplay smoothness
- Developed procedural terrain generation to ensure replayability and variety in player experience

TECHNICAL SKILLS

Languages: Python, C/C++, SML, HTML, CSS, JavaScript, Java

Technologies: Git, GitHub, Linux, Unix, Bazel, Gerrit, CUDA, OpenMP